

Project Report
UPI LEAK DETECTOR

Automated AI-Powered Financial Tracking System

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1. Introduction

UPI Leak Detector is a smart financial application designed to help users track their UPI and bank expenses effortlessly. In the modern digital economy, keeping track of numerous small UPI transactions is challenging. By leveraging Artificial Intelligence, the application automatically extracts transaction data from bank statements (PDFs), receipts (Images), and CSV files. It categorizes spending and identifies hidden 'money leaks' to help users maintain a healthy budget and improve their financial health.

2. Core Features

- AI-Powered Statement Parsing: Uses Google Gemini 2.5 Flash AI to intelligently extract and format debits from complex bank statements.
- Universal Regex Fallback: A robust custom regex engine that works offline to extract dates, merchants, and amounts from any bank PDF.
- Automated Categorization: Sorts transactions into predefined categories (Food, Shopping, Transport, etc.) using keyword heuristics and AI.
- Receipt Scanner: Users can upload an image of a bill, and the AI extracts the merchant and amount instantly.
- Spend Leak Detection: Automatically flags frequent small transactions at the same merchant and unusually large expenses.
- AI Financial Advisor: A built-in, context-aware chatbot that provides personalized financial advice based on the user's recent spending history.
- PDF Export: Users can generate customizable PDF reports of their monthly spending directly from the dashboard.
- Bank-Grade Security: App handles encrypted PDFs safely without storing user passwords. Passwords for user accounts are hashed using bcrypt.

3. Technology Stack

- Backend Framework: Python, Flask
- Database: PostgreSQL (Relational Database)
- Frontend: HTML5, CSS3, JavaScript, Bootstrap 5 (Glassmorphism UI Design)
- AI Engine: Google Gemini GenAI API
- Data Extraction: pdfplumber, pandas, Python re (Regex)

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- Authentication: Flask-Login, Flask-Bcrypt
- PDF Generation: FPDF

4. System Architecture

The application follows a standard Client-Server architecture. The frontend, built with responsive Bootstrap 5 and a premium dark/light mode toggle, communicates with the Flask backend. The backend manages user sessions via Flask-Login. When a user uploads a statement, the backend routes the file to the parsing module. If an API key is available, the AI parser is used; otherwise, the system falls back to the robust Regex parser. Data is structured and safely stored in PostgreSQL. The dashboard securely queries this database to present dynamic analytics, charts, and financial alerts.

5. Implementation Details

The core logic resides in 'app.py'. The upload module securely parses multiple file types. PDFs are decoded using 'pdfplumber', and the extracted text is processed. If the PDF is encrypted, the application uses the user-provided password strictly in-memory. The AI chatbot is implemented as a sticky widget that retains conversation context using the user's 50 most recent transactions. The Leak Detector runs an algorithmic scan during dashboard load, grouping transactions by merchant to calculate frequency and average ticket size to flag anomalies.

6. Future Enhancements

- Multi-bank Aggregation: Integrating with the Account Aggregator Framework for real-time syncing.
- Predictive Budgeting: Using machine learning on historical data to predict next month's expenses.
- SMS Parsing: Creating a companion Android app to parse SMS for real-time transaction updates.
- Cloud Deployment: Containerizing the application with Docker for seamless AWS/GCP deployment.

7. Conclusion

UPI Leak Detector successfully bridges the gap between raw financial data and actionable insights. By automating the tedious process of expense tracking and utilizing advanced AI for categorization and personalized advice, the project provides a comprehensive, secure, and user-friendly personal finance

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management tool.